

Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

1 Plants contain starch and cellulose.

(a) (i) Which part of a plant cell stores starch?

(1)

- ☐ **A** amyloplast
- ☐ **B** middle lamella
- ☐ **C** plasmodesmata
- ☐ **D** tonoplast

(ii) How many of the following statements about starch are correct?

1. it has a compact shape
2. it contains 1,6 glycosidic bonds only
3. it is a polymer of β -glucose
4. it is a polypeptide

(1)

- ☐ **A** one
- ☐ **B** two
- ☐ **C** three
- ☐ **D** four

(iii) How many of the following contain cellulose?

1. phloem
2. sclerenchyma
3. vacuole
4. xylem

(1)

- ☐ **A** one
- ☐ **B** two
- ☐ **C** three
- ☐ **D** four

DO NOT WRITE IN THIS AREA

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(b) Microfibrils are composed of cellulose.

The photograph shows the arrangement of microfibrils in a plant cell wall, as seen using an electron microscope.



(Source: © Biophoto associates/Science photo library)

Explain how the structures of cellulose and microfibrils increase the strength of a plant cell wall.

Use the information in the photograph to support your answer.

(3)

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(Total for Question 1 = 6 marks)

